Western Governor's University

D208 Predictive Modeling

Task 2: Logistic Regression Modeling

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4/10/2025

A. Describe the purpose of this data analysis by doing the following in a document file:

1. Summarize **one** research question that is relevant to a real-world organizational situation captured in the data set you have selected and that you will answer using logistic regression.

Which variables have the most significant effect on whether or not a patient will be re-admitted after initial treatment?

2. Define the goals of the data analysis.

This data analysis aims to build a refined logistic model able to determine whether a patient will need to be readmitted by selecting the most significant independent variables from the data set from an initial logistic model.

Part II: Method Justification

B. Describe logistic regression methods by doing the following in a document file:

1. Summarize **four** assumptions of a logistic regression model.

Logistic regression assumes that the model's dependent variable is binary. It also assumes that there is a linear relationship between the explanatory variables and the logit function. Logistic regression also assumes no multicollinearity between the independent variables. Finally, it is assumed that all observations from our medical dataset are independent of each other.

((Bobbitt,2020)

2. Describe **two** benefits of using Python or R in support of various phases of the analysis.

Python will be used for this data analysis because of the packages available, such as seaborn and matplotlib for visualization. Additionally, the ease of panda's capability to transform the data file into a dataframe and work with the data is extremely helpful, particularly for building and operating within dummy dataframes.

3. Explain why logistic regression is an appropriate technique to analyze the research question summarized in part I.

Logistic regression will be the appropriate technique because the dependent variable to be tested is a binary categorical variable and the goal of the data analysis is a binary yes/no recommendation.

(Bobbitt,2020) (Sewell)

Part III: Data Preparation

C. Summarize the data preparation process for logistic regression by doing the following in a document file:

1. Describe your data cleaning goals and the steps used to clean the data to achieve the goals that align with your research question, including the annotated code.

First, the data has many yes/no columns. Data analysis will be much easier if these are converted to 1/0. This is achieved by running a .replace function over the entire dataframe converting yes/no to 1/0. This is done after creating a dummies dataframe to ensure to the dummy values are also one hot encoded.

Duplicates will be evaluated using .duplicate function, but no fields populate that are necessary to change.

Outlier analysis is also performed on all numeric columns to be used in the initial logistic model, and outliers are replaced with the column mean. There are no columns with outliers however, so no changes are made beyond the fill NA function added as a just in case. (Kavisekar, 2020)

All categorical columns are evaluated using .unique to ensure there are no duplicates, nulls, or values in need of conversion, and these all look good to proceed.

(Sewell)

See annotated code: “D208Proj2BJV.py”

2. Describe the dependent variable and all independent variables using summary statistics that are required to answer the research question, including a screenshot of the summary statistics output for each of these variables.

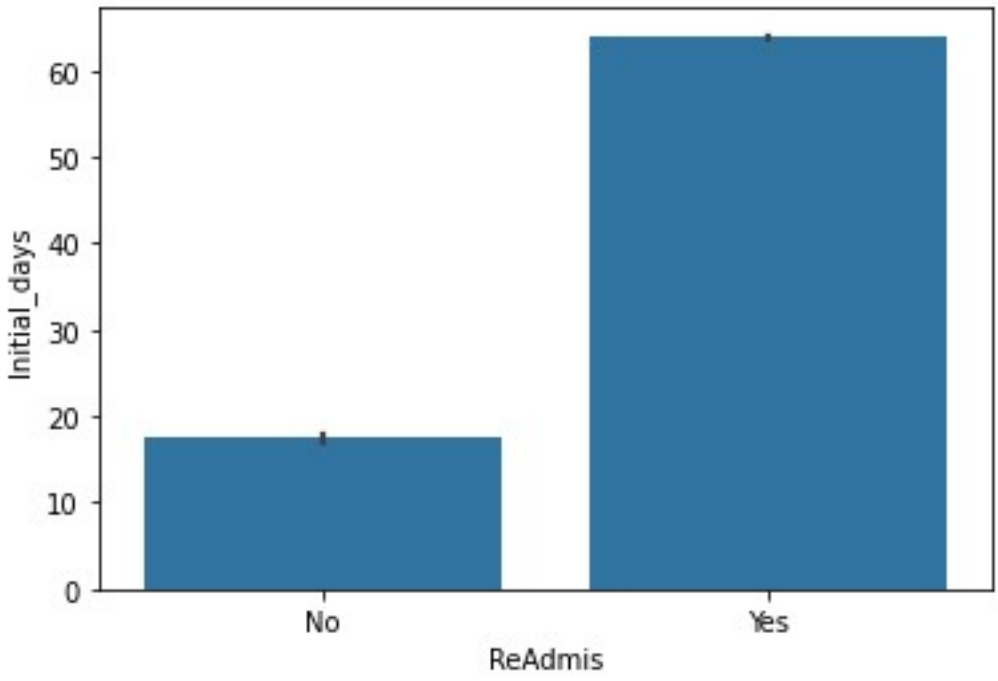
The summary statistics show that readmission is most often not necessary, with 6331 non re-admitted patient interactions out of the 10,000 total interactions. The average patient is 53.5 years old with a max of 89 years. There are 5018 females in the dataset. Emergency is the most common admission reason with 5060 such occurrences. Most patients marked that they do not experience anxiety, at 6785 such responses. The average patient stays 34.4 days in the hospital, and the longest stay is just under 72 full days. We are able to see that 58.65% of patients do not have reflux esophagitis. The most commonly used service is bloodwork, with 5265 patients needing it. 66.28% of patients do not have hyperlipidemia. 72.62% of patients are free of diabetes. 70.94% of patients do not classify as overweight. 80.07% of patients have not experienced a stroke prior to admission. 59.10% of patients do not have high blood pressure.

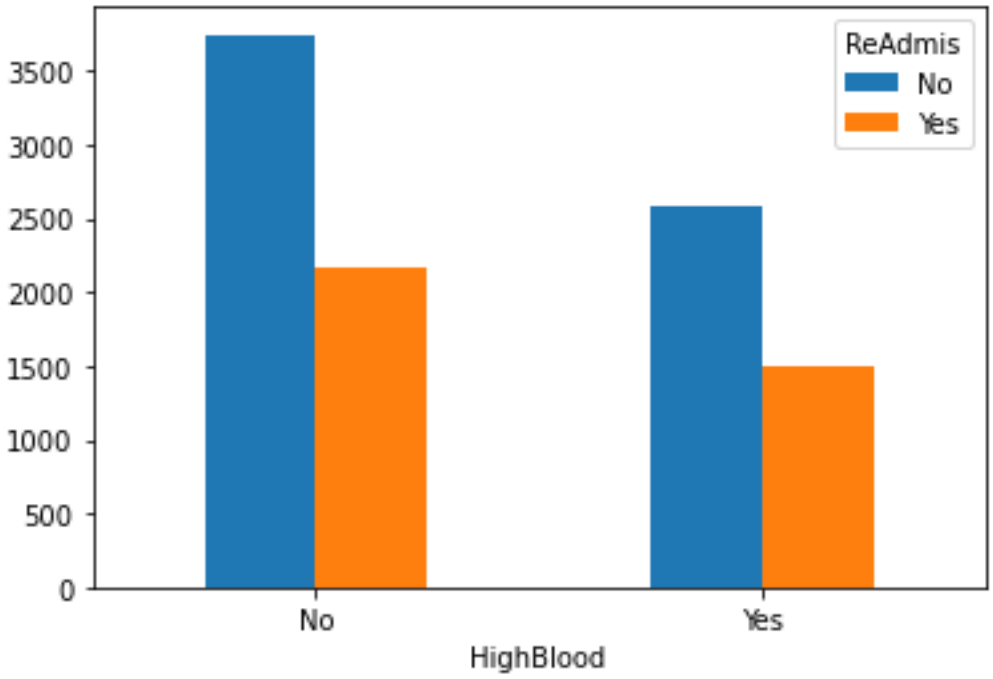
```
Name: count, dtype: int64
count     10000
unique        2
top          No
freq       6331
Name: ReAdmis, dtype: object
count    10000.000000
mean        53.511700
std         20.638538
min         18.000000
25%         36.000000
50%         53.000000
75%         71.000000
max         89.000000
Name: Age, dtype: float64
count      10000
unique         3
top       Female
freq        5018
Name: Gender, dtype: object
count                   10000
unique                      3
top       Emergency Admission
freq                     5060
Name: Initial_admin, dtype: object
count     10000
unique        2
top          No
freq       6785
Name: Anxiety, dtype: object
```

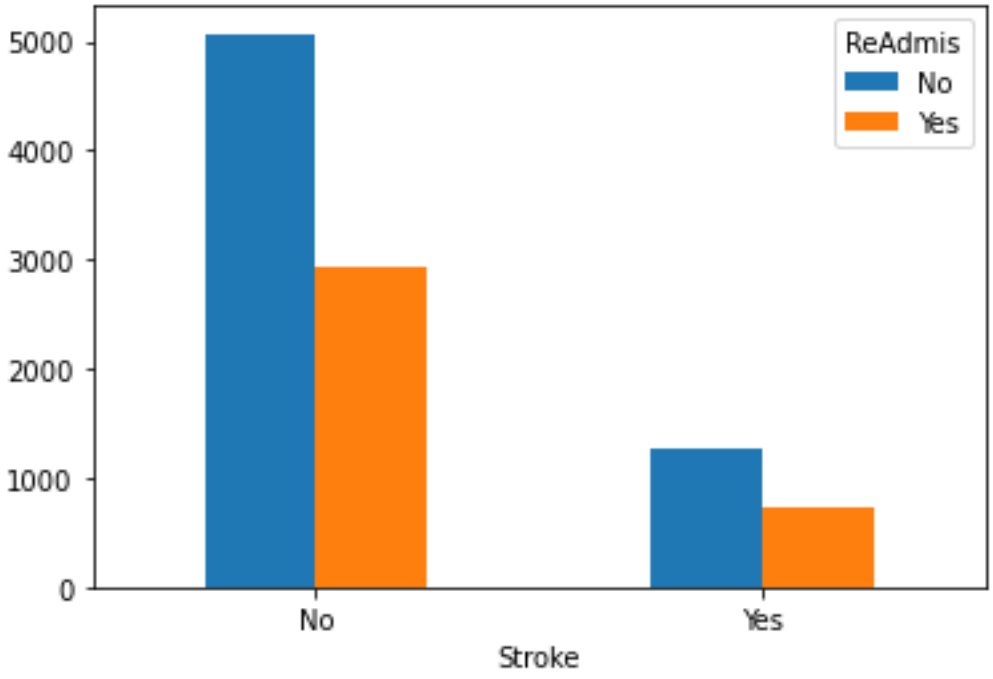
```
count    10000.000000
mean        34.455299
std         26.309341
min          1.001981
25%          7.896215
50%         35.836244
75%         61.161020
max         71.981490
Name: Initial_days, dtype: float64
count     10000
unique        2
top          No
freq       5865
Name: Reflux_esophagitis, dtype: object
count          10000
unique             4
top       Blood Work
freq            5265
Name: Services, dtype: object
count     10000
unique        2
top          No
freq       6628
Name: Hyperlipidemia, dtype: object
count     10000
unique        2
top          No
freq       7262
Name: Diabetes, dtype: object
count     10000
unique        2
top         Yes
freq       7094
Name: Overweight, dtype: object
```

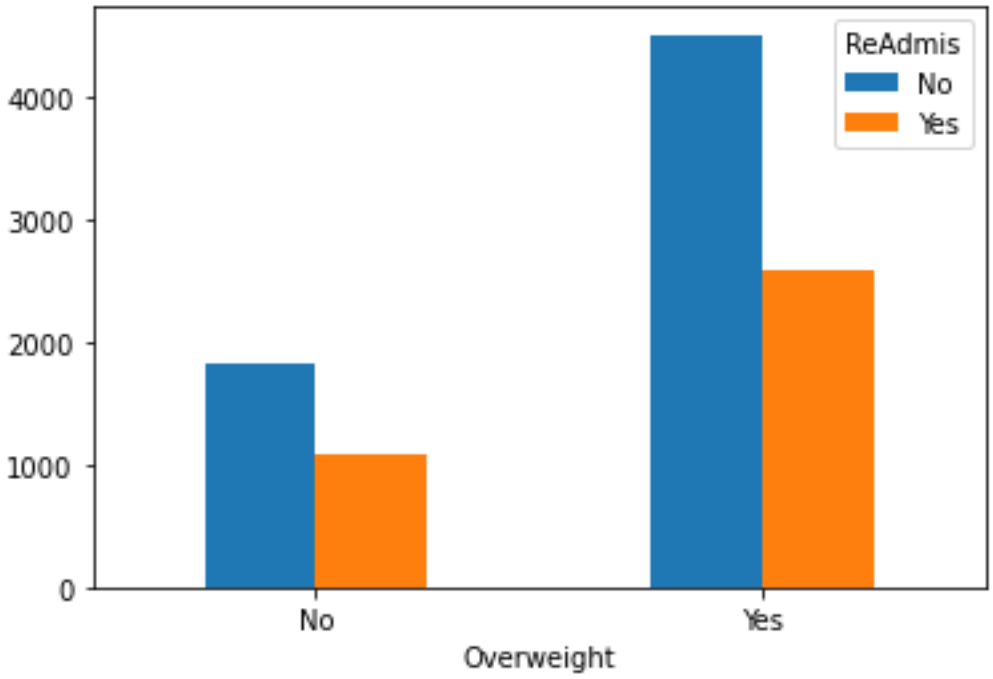
```
count     10000
unique        2
top          No
freq       8007
Name: Stroke, dtype: object
count     10000
unique        2
top          No
freq       5910
Name: HighBlood, dtype: object
```

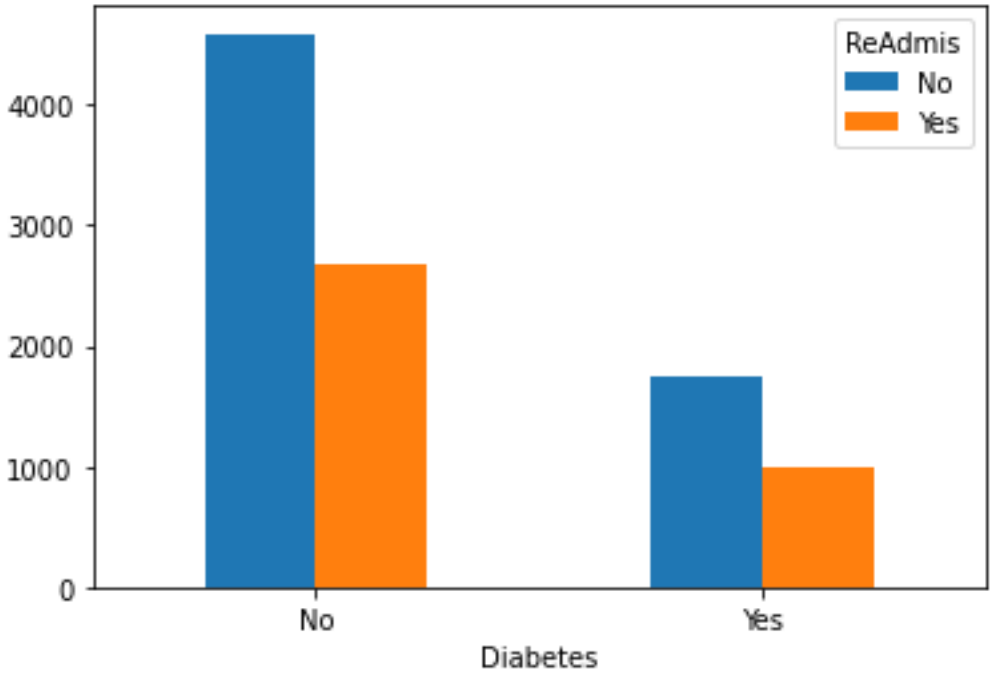
3. Generate univariate and bivariate visualizations of the distributions of the dependent and independent variables, including the dependent variable in your bivariate visualizations.

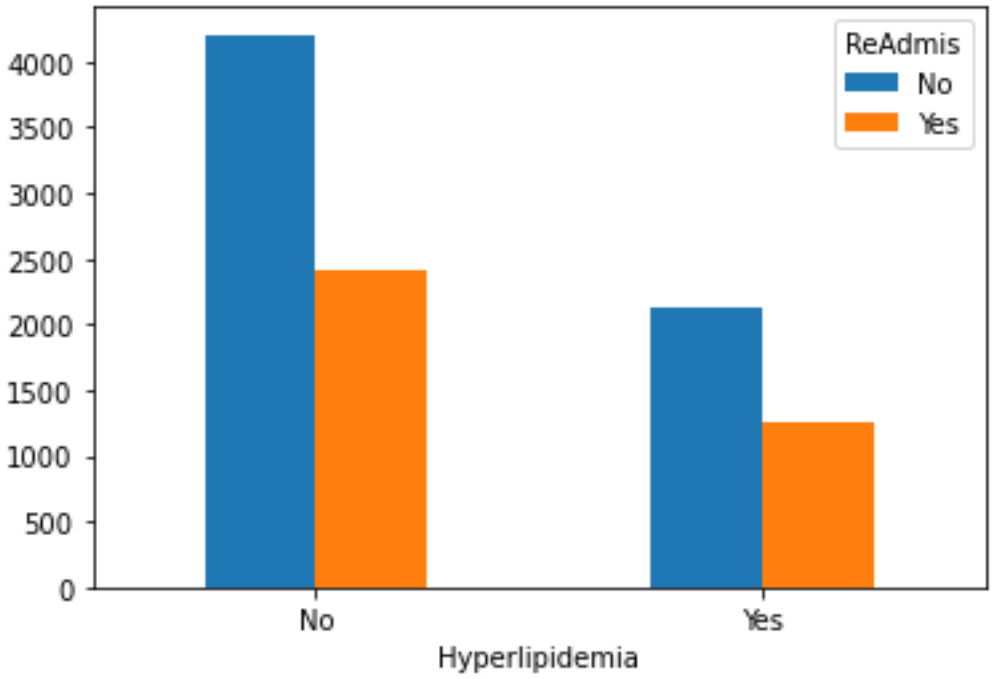

60
50
40
30
20
10
0
Initial_days
No
Yes
ReAdmis

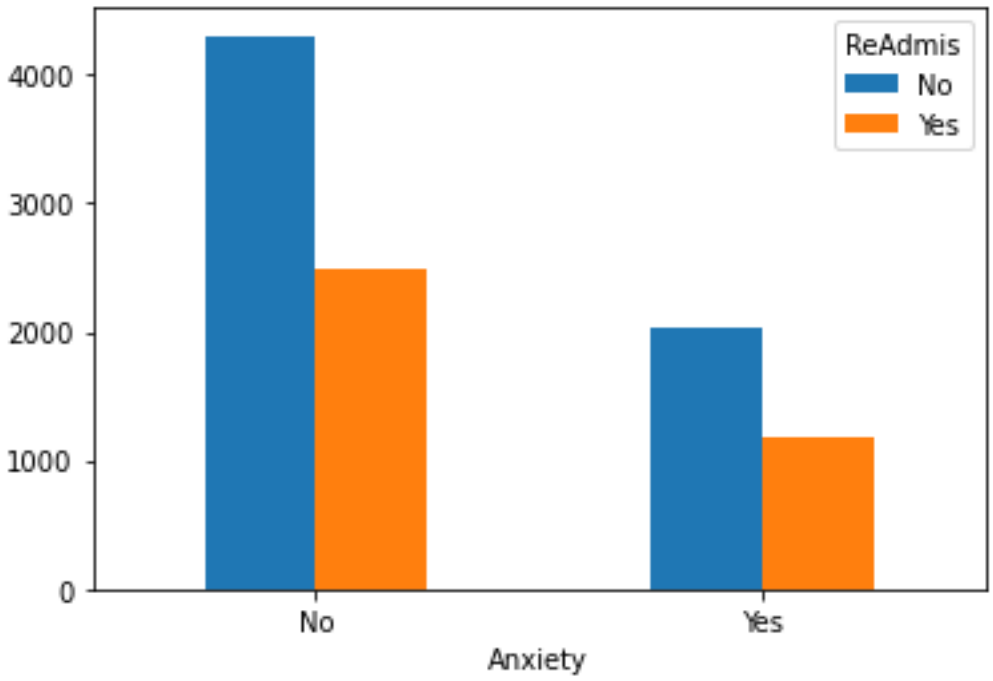

3500
3000
2500
2000
1500
1000
500
0
ReAdmis
No
Yes
No
Yes
HighBlood

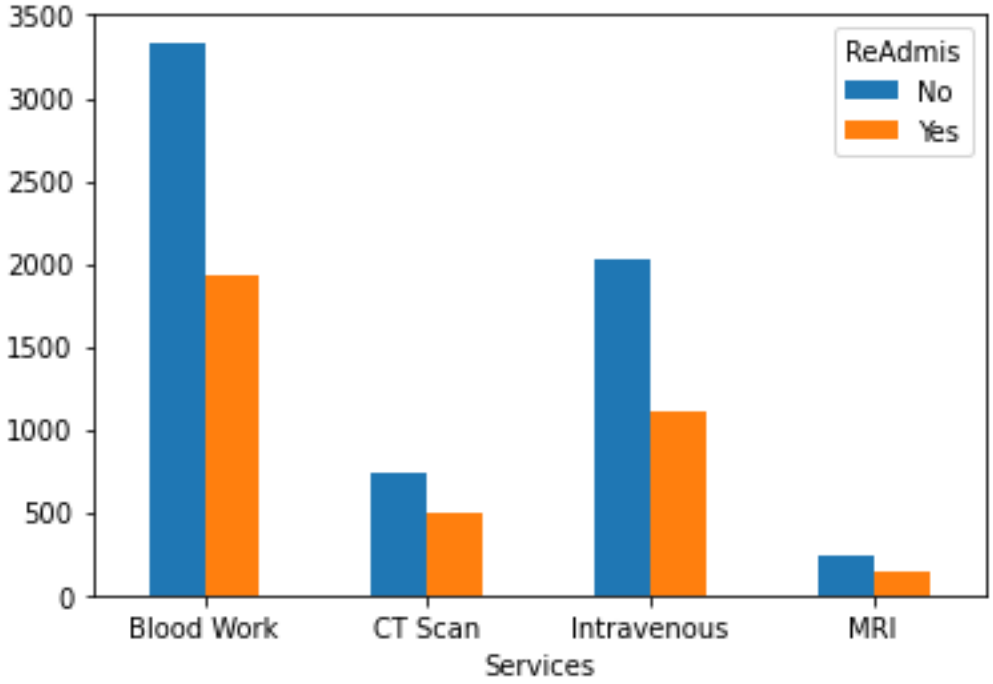

ReAdmis
No
Yes
5000
4000
3000
2000
1000
0
No
Yes
Stroke

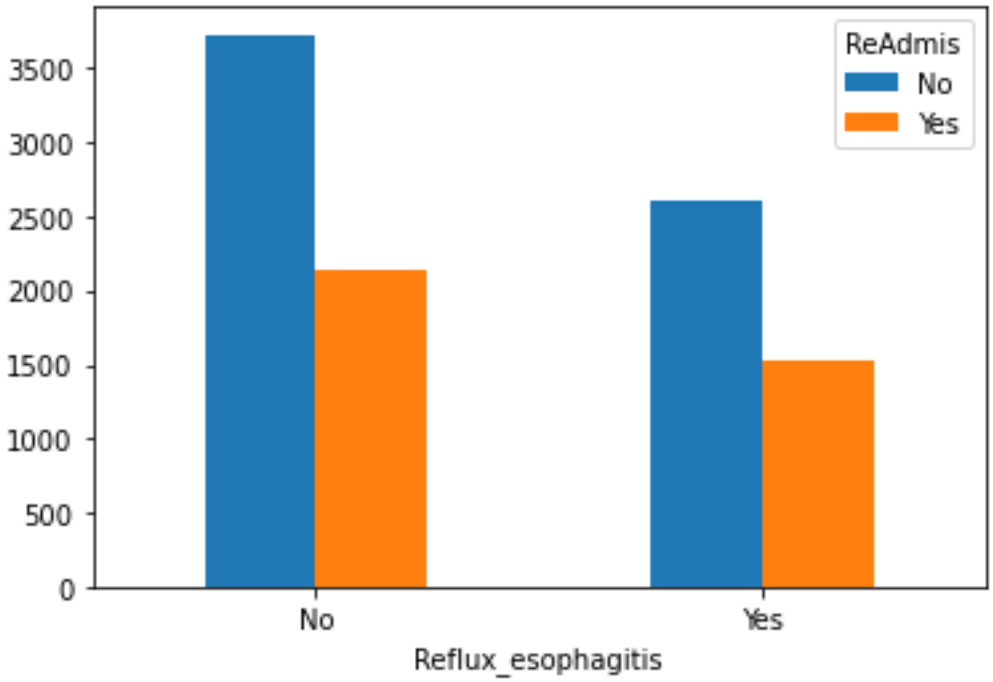

ReAdmis
No
Yes
4000
3000
2000
1000
0
No
Yes
Overweight

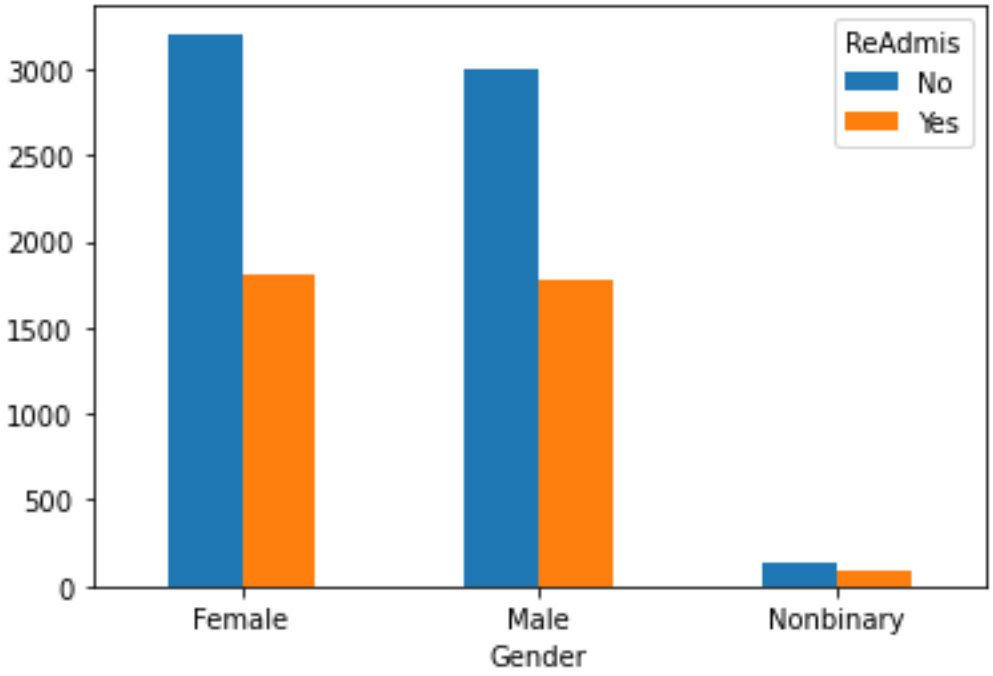

ReAdmis
No
Yes
4000
3000
2000
1000
0
No
Yes
Diabetes

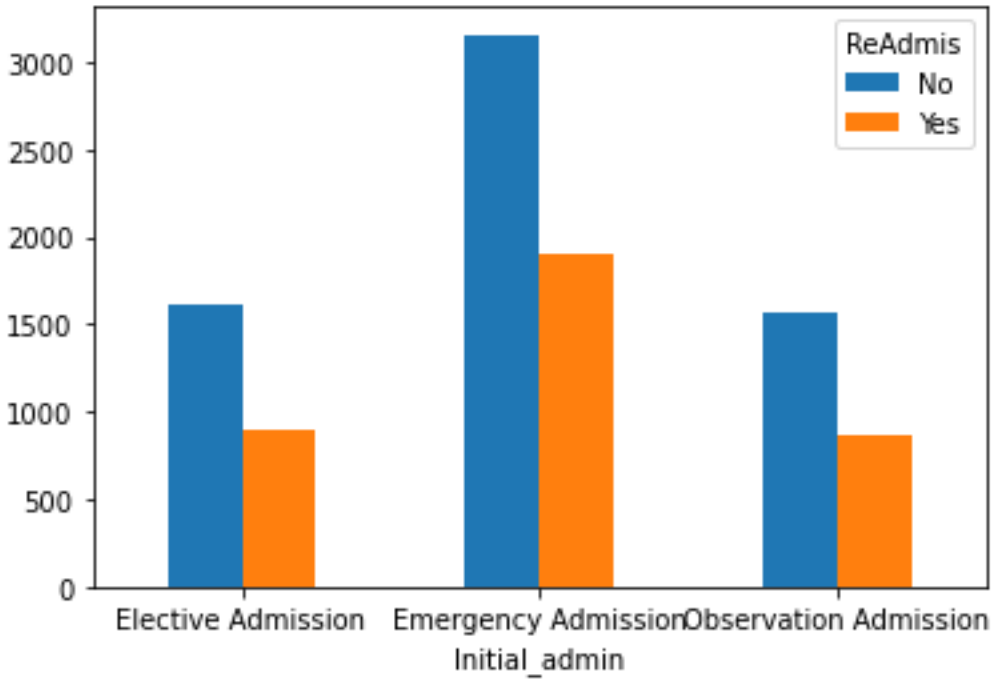

ReAdmis
No
Yes
4000
3500
3000
2500
2000
1500
1000
500
0
No
Yes
Hyperlipidemia

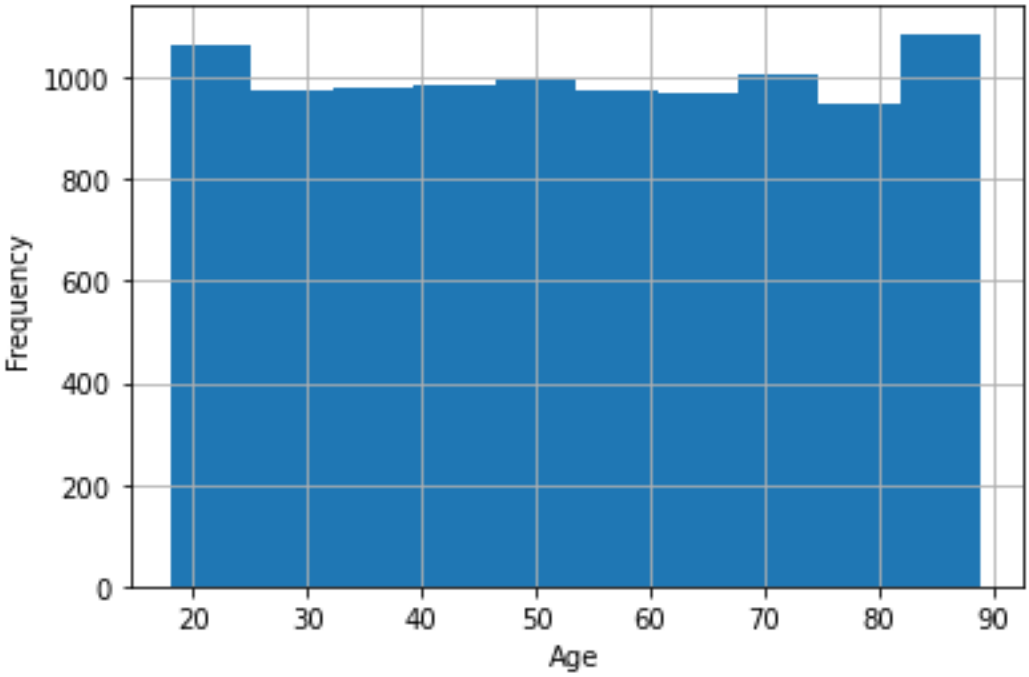

ReAdmis
No
Yes
4000
3000
2000
1000
0
No
Yes
Anxiety

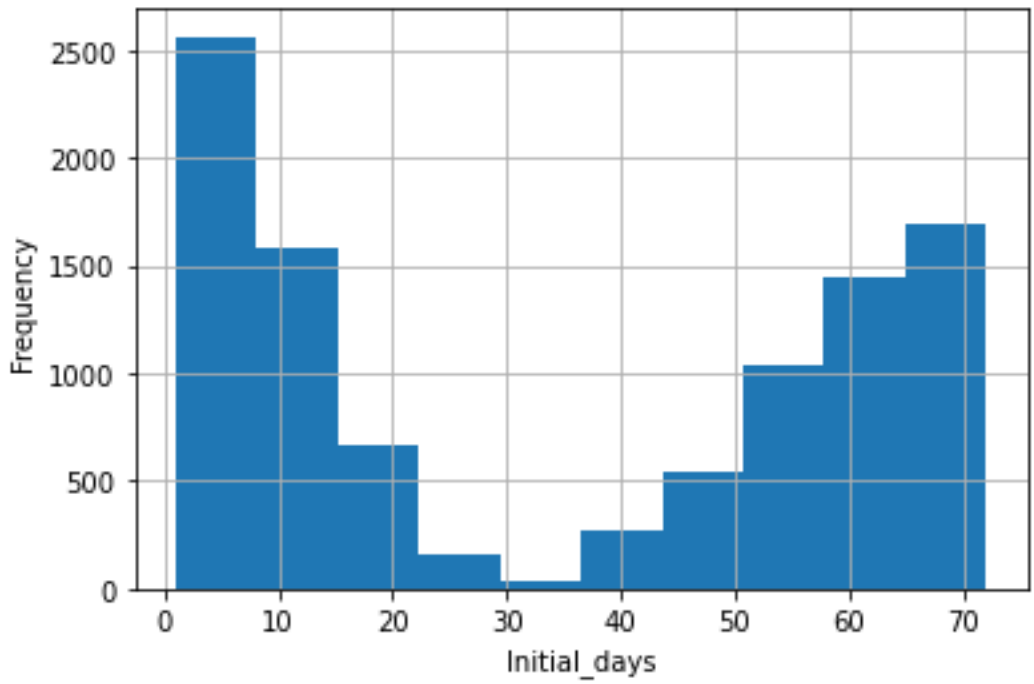

ReAdmis
No
Yes
3500
3000
2500
2000
1500
1000
500
0
Blood Work
CT Scan
Intravenous
MRI
Services


ReAdmis
No
Yes
3500
3000
2500
2000
1500
1000
500
0
No
Yes
Reflux_esophagitis


ReAdmis
No
Yes
3000
2500
2000
1500
1000
500
0
Female
Male
Nonbinary
Gender


ReAdmis
No
Yes
3000
2500
2000
1500
1000
500
0
Elective Admission
Emergency Admission
Observation Admission
Initial_admin


1000
800
600
400
200
0
Frequency
20
30
40
50
60
70
80
90
Age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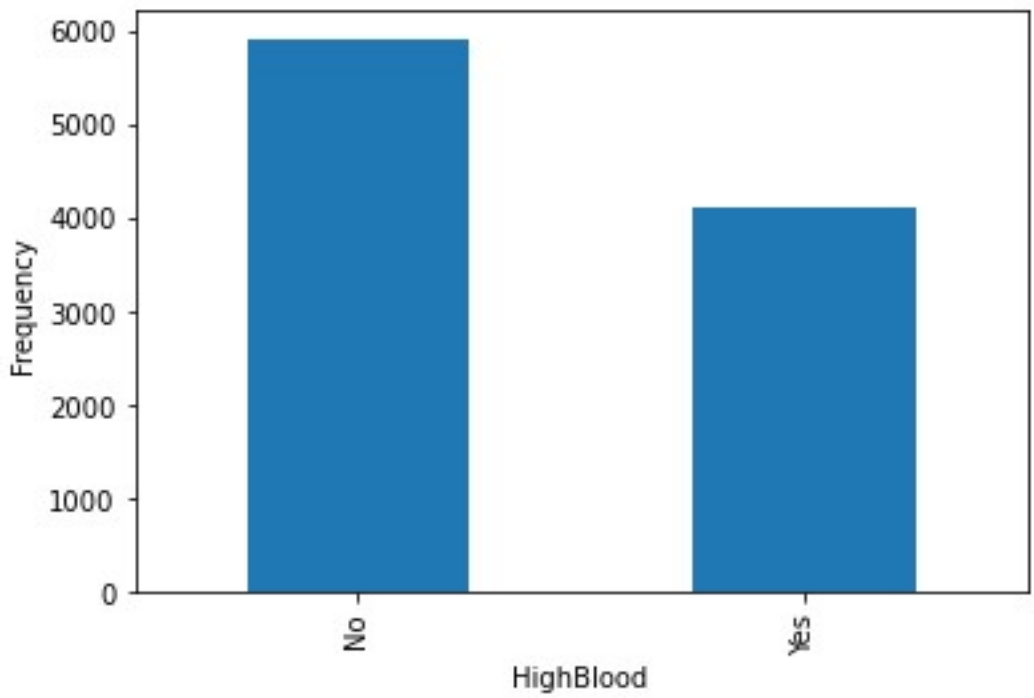
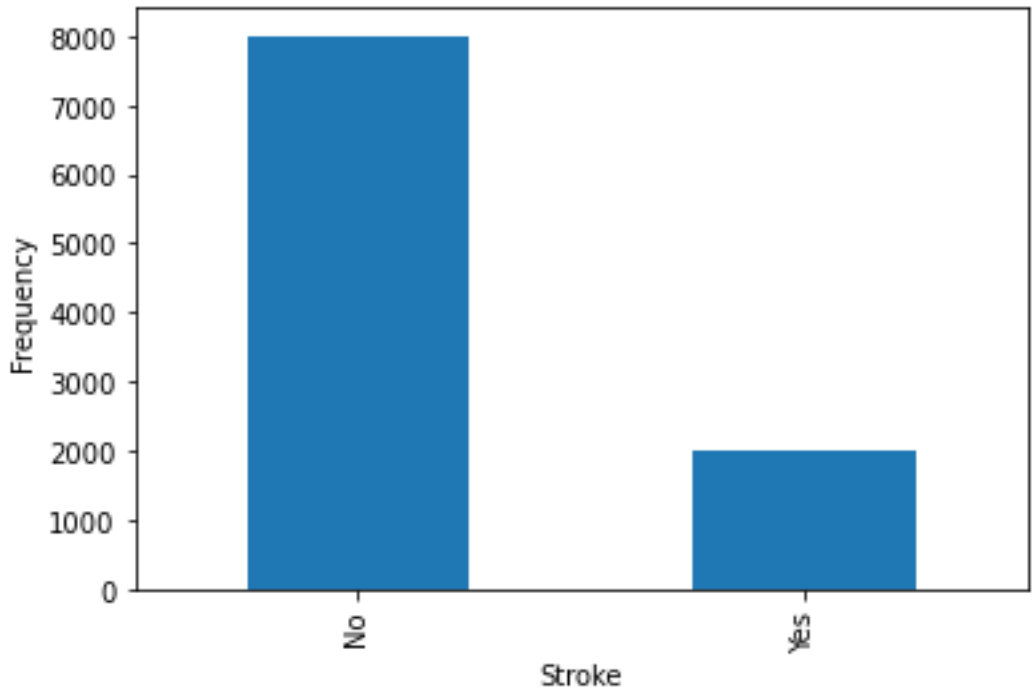
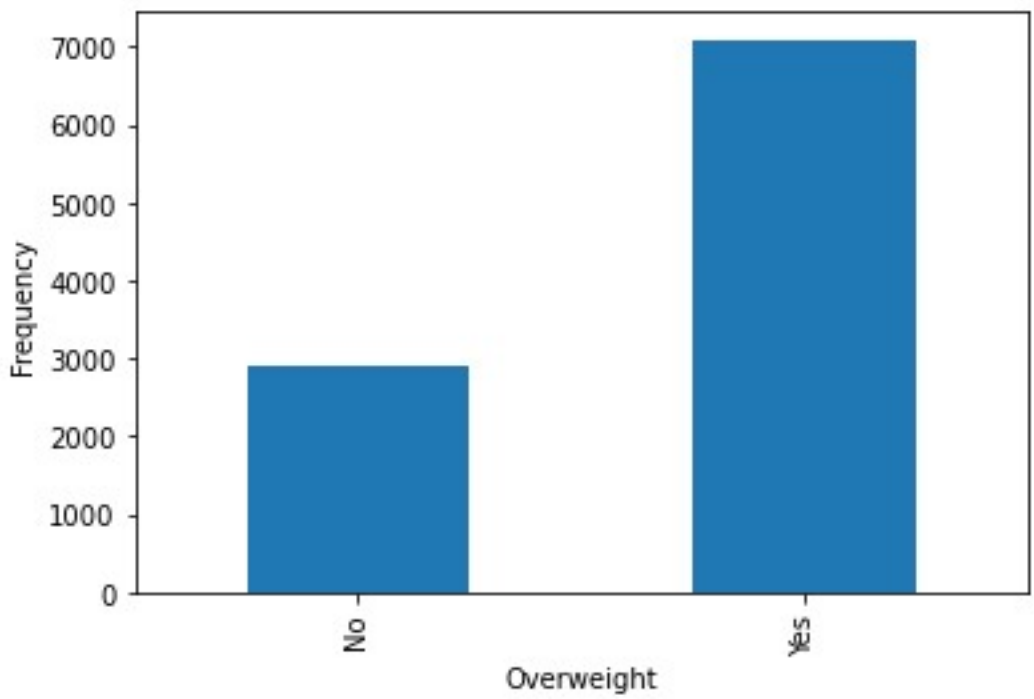
2000
1500
1000
500
0
Frequency
0
10
20
30
40
50
60
70
Initial_days


6000
5000
4000
3000
2000
1000
0
Frequency
No
Yes
HighBlood


8000
7000
6000
5000
4000
3000
2000
1000
0
Frequency
No
Yes
Stroke


7000
6000
5000
4000
3000
2000
1000
0
Frequency
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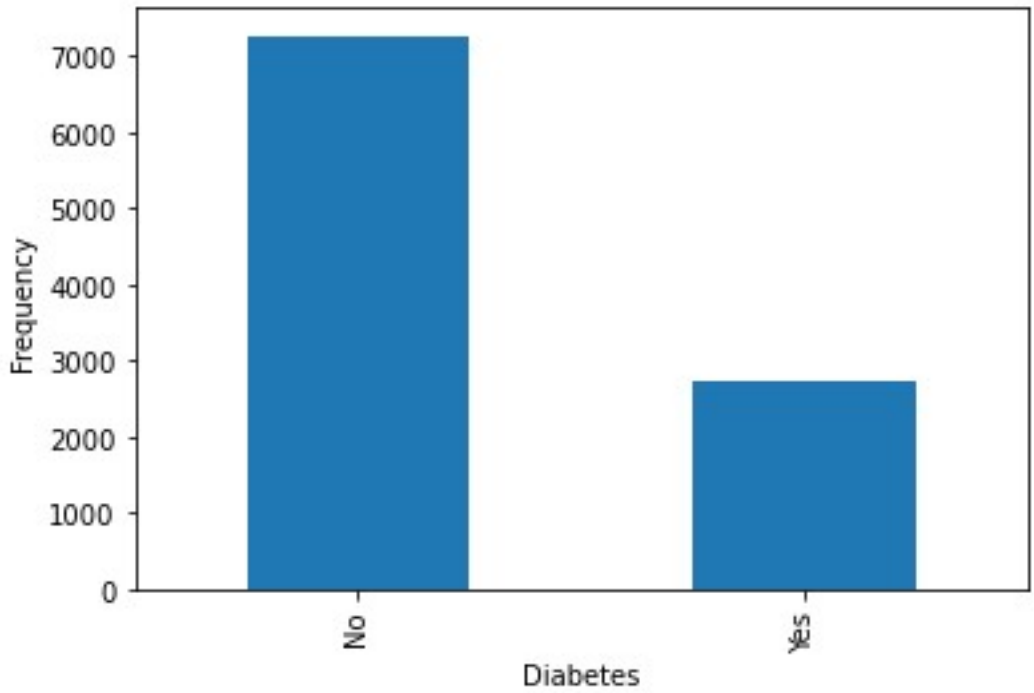
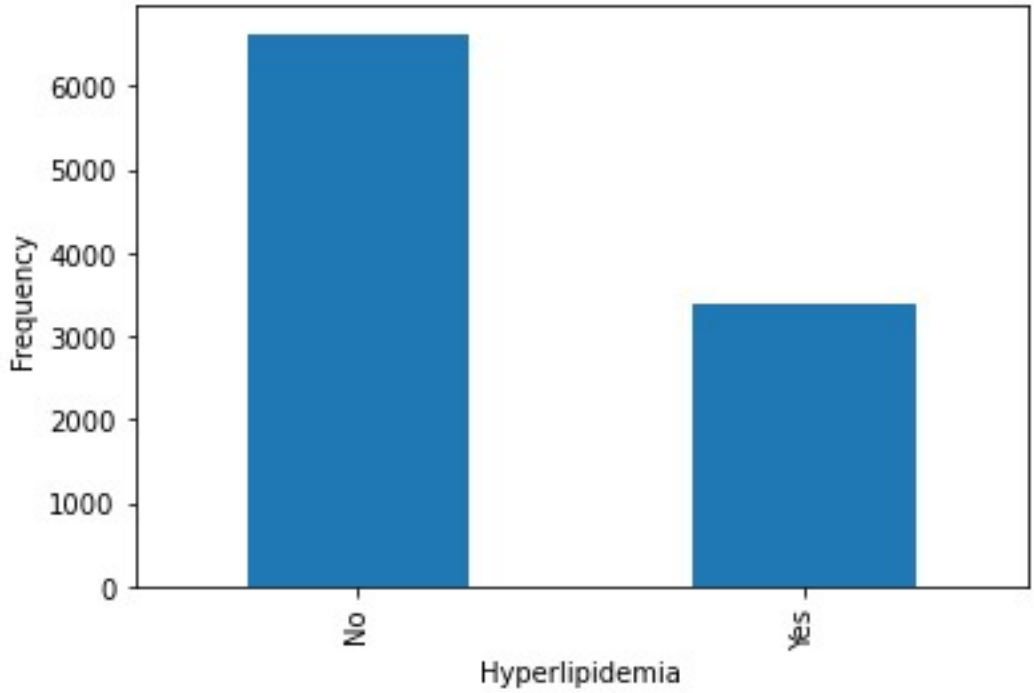
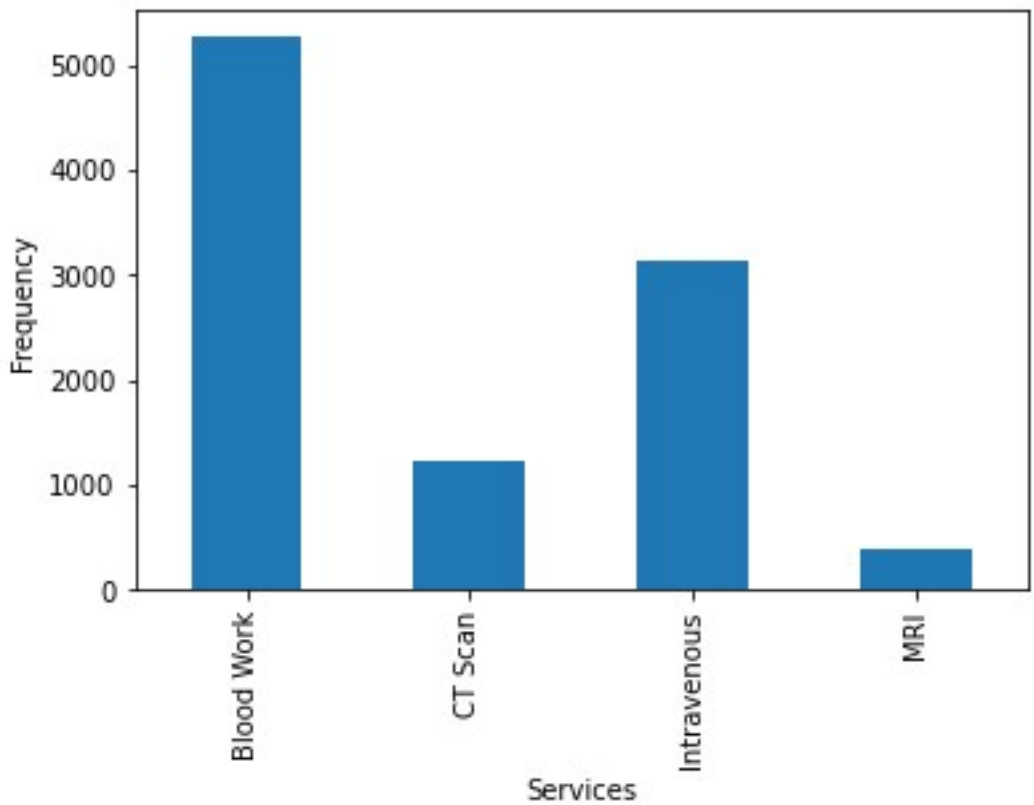
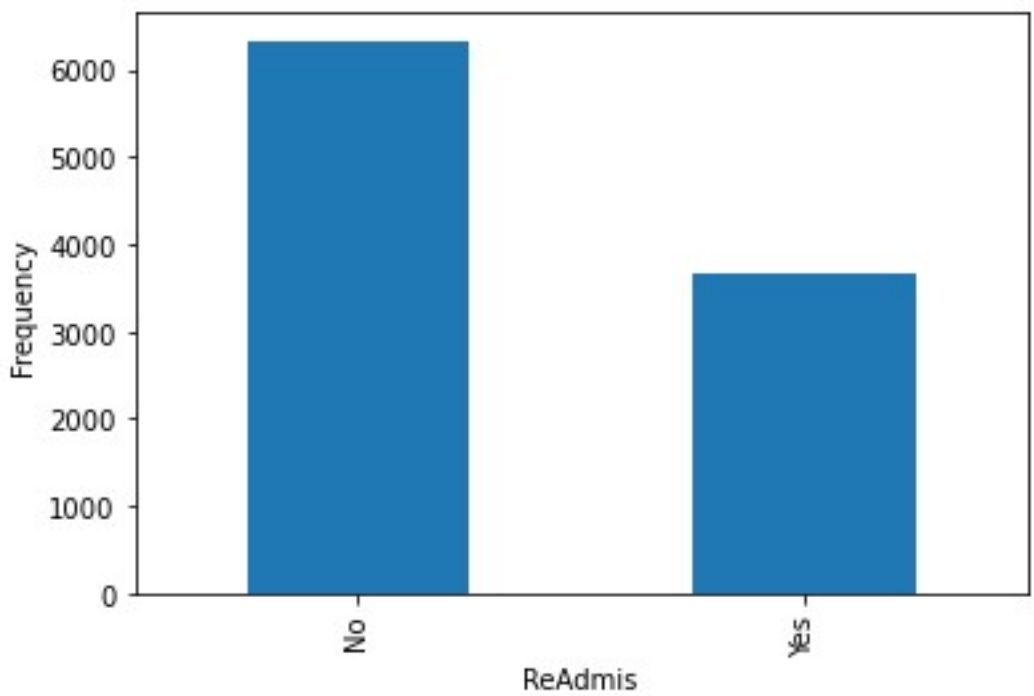
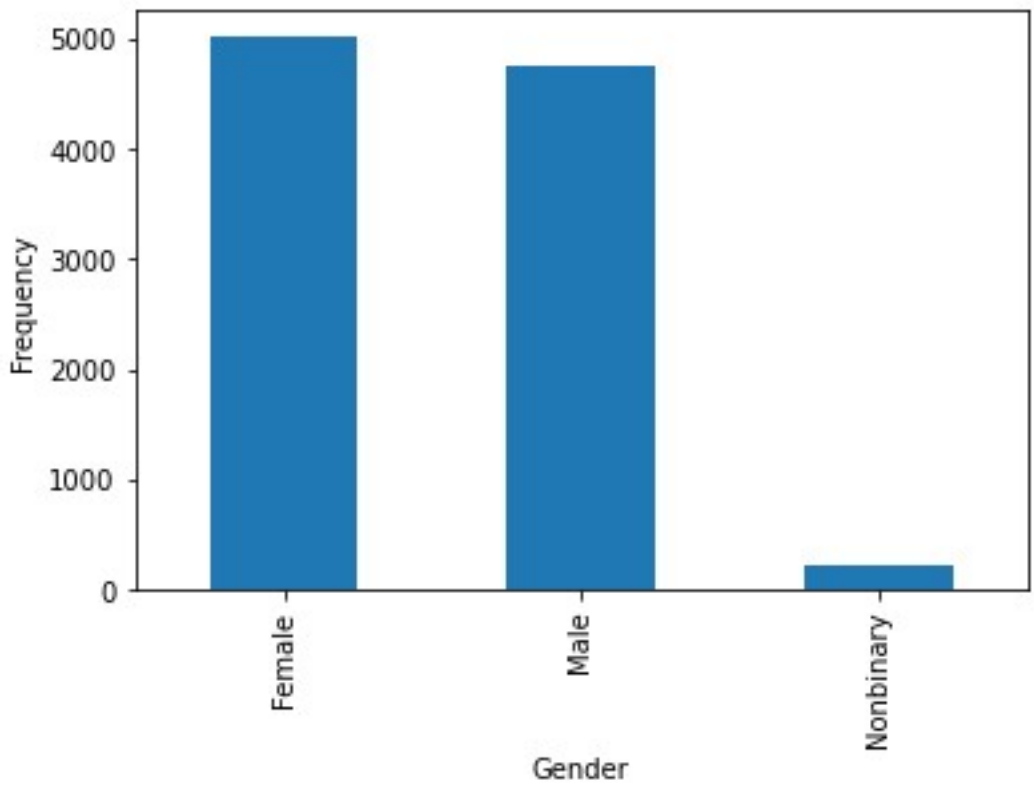
Yes
Overweight

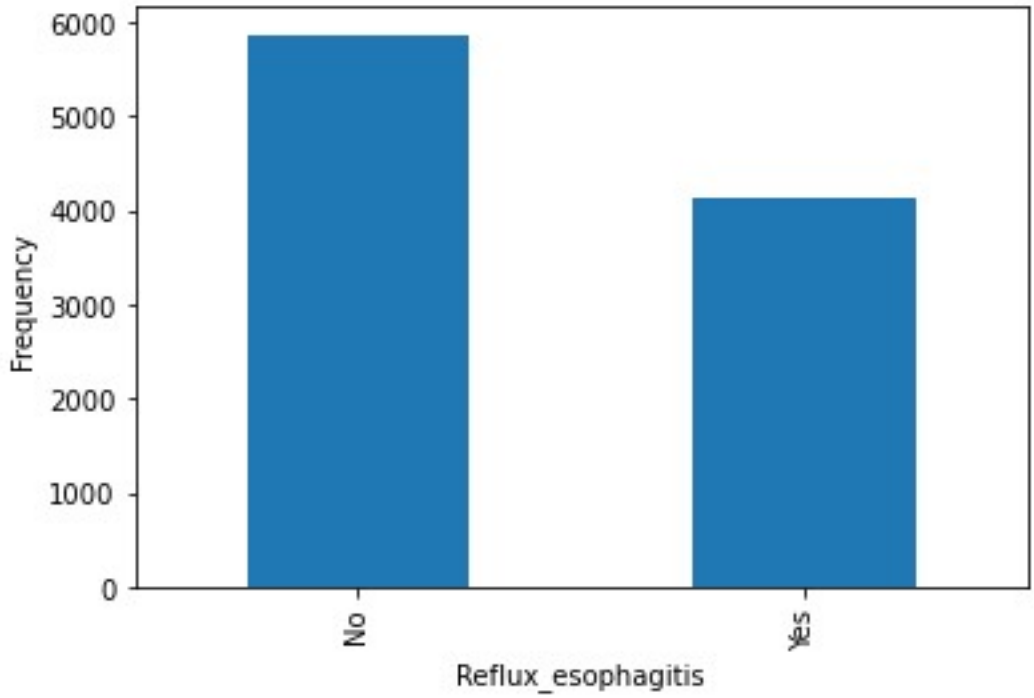

7000
6000
5000
4000
3000
2000
1000
0
Frequency
No
Yes
Diabetes

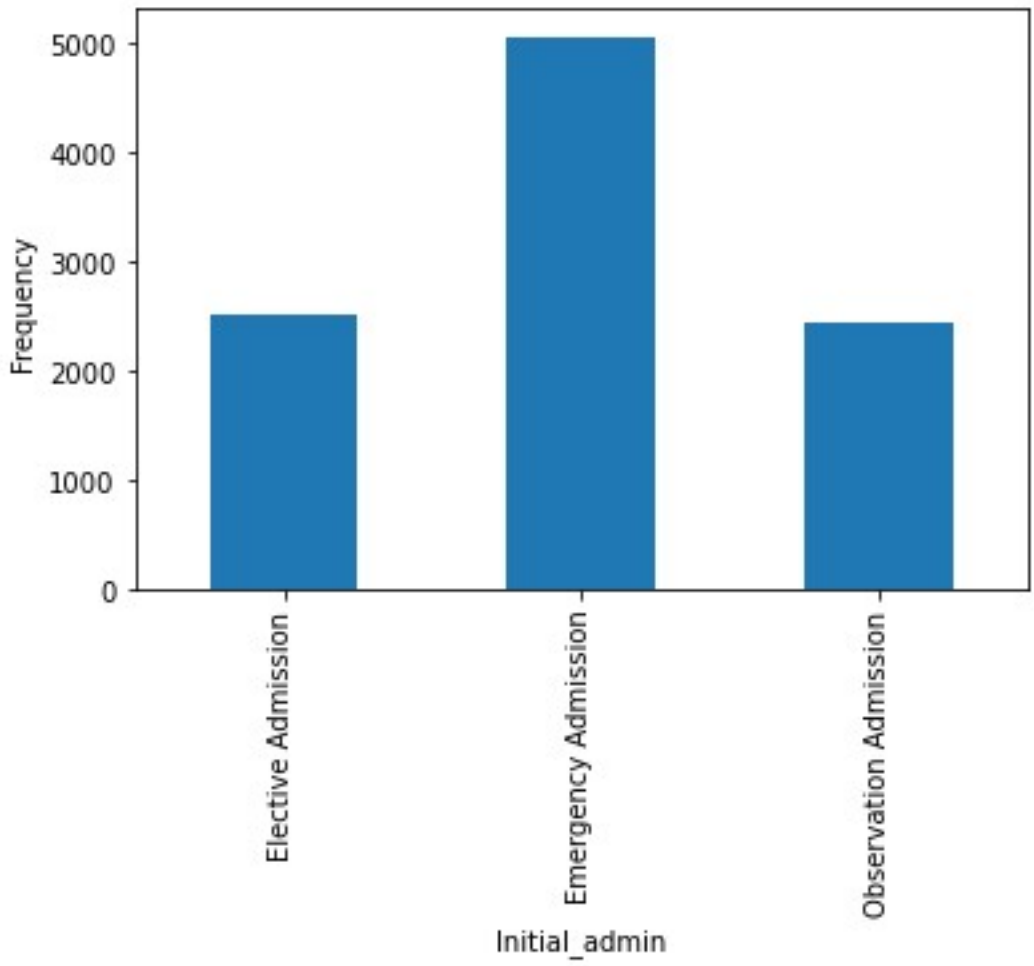

6000
5000
4000
3000
2000
1000
0
Frequency
No
Yes
Hyperlipidemia

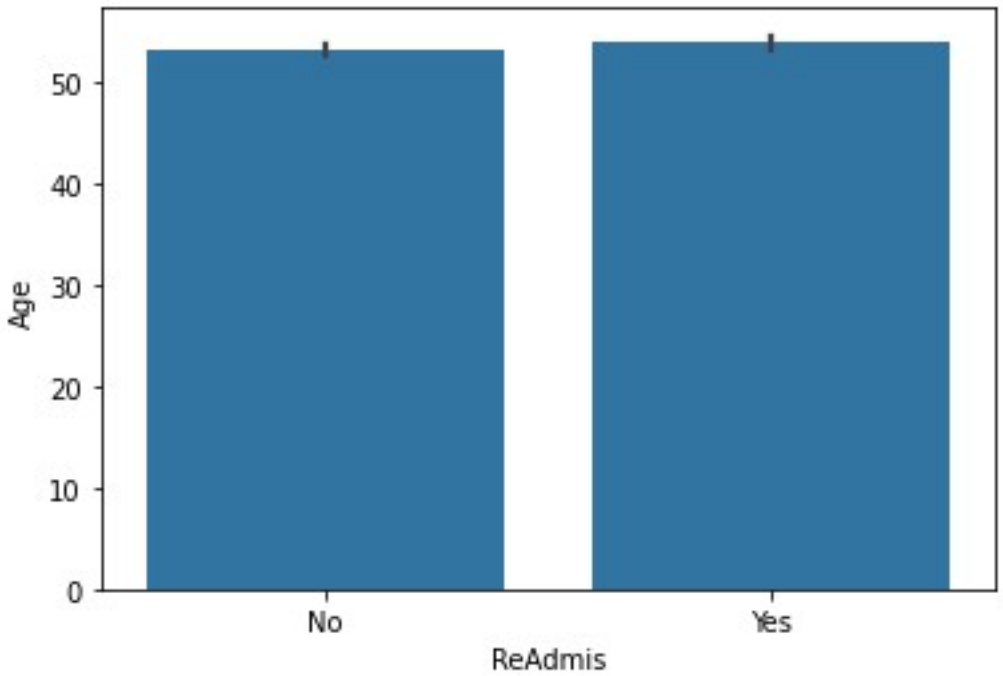

5000
4000
3000
2000
1000
0
Frequency
Blood Work
CT Scan
Intravenous
MRI
Services

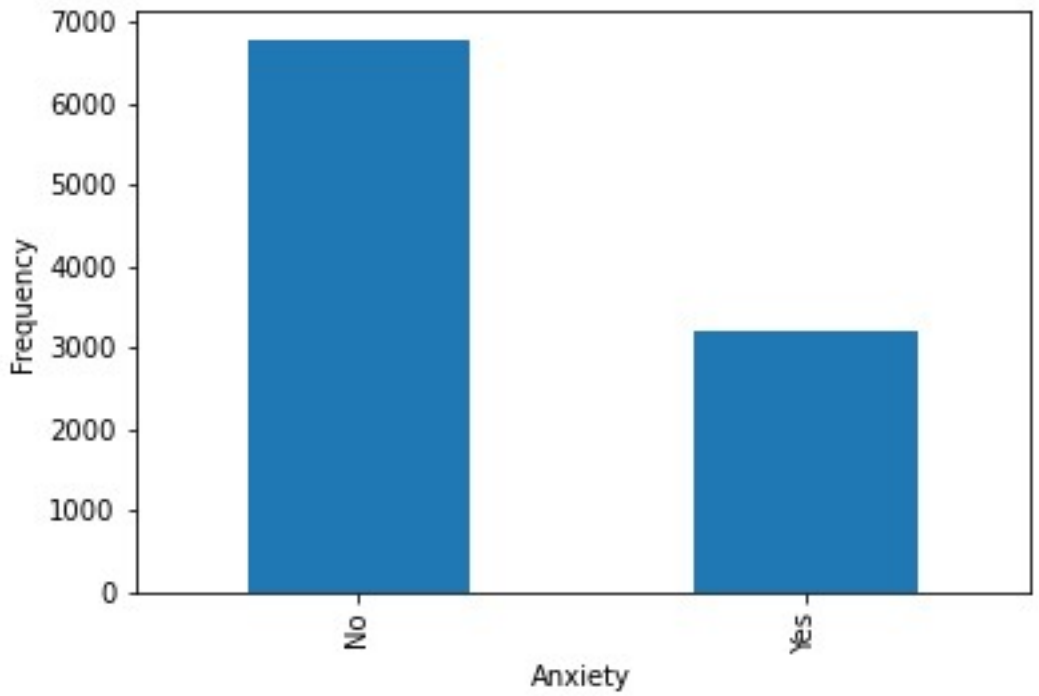

6000
5000
4000
3000
2000
1000
0
Frequency
No
Yes
ReAdmis


5000
4000
3000
2000
1000
0
Frequency
Female
Male
Nonbinary
Gender


6000
5000
4000
3000
2000
1000
0
Frequency
No
Yes
Reflux_esophagitis


5000
4000
3000
2000
1000
0
Frequency
Elective Admission
Emergency Admission
Observation Admission
Initial_admin





(Kumar,2022) (GeeksforGeeks,2021)

4. Describe your data transformation goals that align with your research question and the steps used to transform the data to achieve the goals, including the annotated code.

The categorical columns will need to be transformed into dummy columns and built into a new dataframe. This will be done using pandas' get dummies function and assigning these values to a dataframe that will be evaluated in feature selection.

("Multiple Linear Regression — Basic Analytics in Python") (Kavisekar, 2020)

The dummy variables will need to be one hot encoded, converting any True/False and Yes/No columns into 1/0s so we can run calculations.

The data will then be split into a training and test group, using an 80/20 split.

Then, the data will be normalized using the minmaxscaler.

Finally, backwards RFE will be implemented on the logistic model variables until ten features remain for the reduced model.

(RITHP)

5. Provide the prepared data set as a CSV file.

See file: "InitialLogDummyFeatures.csv"

Part IV: Model Comparison and Analysis

D. Compare an initial and a reduced logistic regression model by doing the following in a document file:

1. Construct an initial logistic regression model from *all* independent variables that were identified in part C2.

```
<class 'statsmodels.iolib.summary.Summary'>
"""
                           Logit Regression Results
==============================================================================
Dep. Variable:                      y   No. Observations:                 8000
Model:                          Logit   Df Residuals:                     7984
Method:                           MLE   Df Model:                           15
Date:                Tue, 15 Apr 2025   Pseudo R-squ.:                  0.6019
Time:                        14:33:15   Log-Likelihood:                -2098.5
converged:                       True   LL-Null:                       -5271.6
Covariance Type:            nonrobust   LLR p-value:                     0.000
==========================================================================================
                                          coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------
Age                                    -2.4447      0.134    -18.267      0.000      -2.707      -2.182
Initial_days                            8.3249      0.184     45.225      0.000       7.964       8.686
 Reflux_esophagitis_Yes                -0.8004      0.080    -10.014      0.000      -0.957      -0.644
Gender_Male                            -0.8199      0.080    -10.307      0.000      -0.976      -0.664
Gender_Nonbinary                       -0.4506      0.267     -1.686      0.092      -0.974       0.073
Anxiety_Yes                            -0.6407      0.084     -7.621      0.000      -0.805      -0.476
Initial_admin_Emergency:Admission      -1.3092      0.096    -13.638      0.000      -1.497      -1.121
Initial_admin_Observation:Admission    -1.5455      0.112    -13.828      0.000      -1.765      -1.326
Services_CT_Scan                       -0.3398      0.128     -2.665      0.008      -0.590      -0.090
Services_Intravenous                   -0.7511      0.088     -8.566      0.000      -0.923      -0.579
Services_MRI                           -0.3161      0.204     -1.550      0.121      -0.716       0.084
Stroke_Yes                             -0.4099      0.097     -4.232      0.000      -0.600      -0.220
Hyperlipidemia_Yes                     -0.5559      0.083     -6.731      0.000      -0.718      -0.394
Diabetes_Yes                           -0.6421      0.088     -7.291      0.000      -0.815      -0.469
Overweight_Yes                         -1.3978      0.089    -15.786      0.000      -1.571      -1.224
HighBlood_Yes                          -0.6005      0.080     -7.512      0.000      -0.757      -0.444
==========================================================================================
"""
```

```
    ...: print(cm)
Initial Y Intercept           : [-17.01035867]
[[3.60713838e-02 9.63928616e-01]
 [9.99990535e-01 9.46454581e-06]
 [9.99999903e-01 9.73228730e-08]
 ...
 [4.16635088e-01 5.83364912e-01]
 [9.99981715e-01 1.82847730e-05]
 [6.00332271e-01 3.99667729e-01]]
              precision    recall  f1-score   support

           0       0.99      0.97      0.98      5040
           1       0.95      0.99      0.97      2960

    accuracy                           0.98      8000
   macro avg       0.97      0.98      0.98      8000
weighted avg       0.98      0.98      0.98      8000

[[4890  150]
 [  25 2935]]
```

(SciKit,2014) (Scikit,2019)

2. Justify a statistically based feature selection procedure or a model evaluation metric to reduce the initial model in a way that aligns with the research question.

Recursive backward feature elimination will be implemented in order to continually analyze the least important feature from the initial model and remove it, repeating continuously until only ten features remain. This will ensure that any changes to the remaining variables are accounted for upon each step of removal, and that will occur as many features are directly related.

(RITHP)

3. Provide a reduced logistic regression model that follows the feature selection or model evaluation process in part D2, including a screenshot of the output for *each* model.

```
<class 'statsmodels.iolib.summary.Summary'>
"""
                           Logit Regression Results
==============================================================================
Dep. Variable:                      y   No. Observations:                 8000
Model:                          Logit   Df Residuals:                     7990
Method:                           MLE   Df Model:                            9
Date:                Tue, 15 Apr 2025   Pseudo R-squ.:                  0.3721
Time:                        13:14:10   Log-Likelihood:                -3309.9
converged:                       True   LL-Null:                       -5271.6
Covariance Type:            nonrobust   LLR p-value:                     0.000
==========================================================================================
                                         coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------
Initial_days                            4.7413      0.102     46.526      0.000       4.542       4.941
Initial_admin_Emergency Admission      -1.2817      0.060    -21.533      0.000      -1.398      -1.165
Stroke_Yes                             -0.9433      0.077    -12.204      0.000      -1.095      -0.792
Services_MRI                           -0.6170      0.163     -3.780      0.000      -0.937      -0.297
Services_CT Scan                       -0.5528      0.095     -5.795      0.000      -0.740      -0.366
Gender_Nonbinary                       -0.5330      0.213     -2.508      0.012      -0.950      -0.117
Anxiety_Yes                            -1.1224      0.066    -16.879      0.000      -1.253      -0.992
HighBlood_Yes                          -1.2295      0.062    -19.696      0.000      -1.352      -1.107
Diabetes_Yes                           -1.0632      0.070    -15.130      0.000      -1.201      -0.925
Hyperlipidemia_Yes                     -1.1044      0.065    -17.062      0.000      -1.231      -0.977
==========================================================================================
"""
```

(Scikit,2014)(Scikit,2019)

E. Analyze the data set using your reduced logistic regression model by doing the following in a document file:

1. Explain your data analysis process by comparing the initial logistic regression model and reduced logistic regression model, including the following element:

- a model evaluation metric

Initial model metrics:

The initial model has a weighted avg precision score of .98, the same as what the reduced model shows. The initial model also has a recall and accuracy score of .98, both within 1% of what the reduced model scores as well.

The reduced model does maintain accuracy, precision and recall, but the Pseudo R square drops from 0.60 to 0.37 – a significant decrease indicating a deprecation of model fit.

The accuracy scores and confusion matrix show that the reduced model will be able to consistently provide accurate predictions, even though the Pseudo R-Square does show that the initial model is a better fitted model to the data.

(Jensen,2025)

2. Provide the output and *all* calculations of the analysis you performed, including the following elements for your reduced logistic regression model:

- confusion matrix
- accuracy calculation

Reduced model metrics:

```
Reduced Y Intercept           : [-16.87657473]
[[3.59422792e-02 9.64057721e-01]
 [9.99991553e-01 8.44719838e-06]
 [9.99999897e-01 1.03293971e-07]
 ...
 [4.02401272e-01 5.97598728e-01]
 [9.99981141e-01 1.88593356e-05]
 [5.95067708e-01 4.04932292e-01]]
              precision    recall  f1-score   support

           0       1.00      0.97      0.98      5040
           1       0.95      0.99      0.97      2960

    accuracy                           0.98      8000
   macro avg       0.97      0.98      0.98      8000
weighted avg       0.98      0.98      0.98      8000

[[4891  149]
 [  23 2937]]
```

The reduced model boasts an accuracy score, a weighted average of recall, and a weighted average of precision, all at 0.98.

3. Provide an executable, error-free copy of the code used to support the implementation of the logistic regression models using a Python or R file.

See annotated code: “D208Proj2BJV.py”

Part V: Data Summary and Implications

F. Summarize your findings and assumptions by doing the following in a document file:

1. Discuss the results of your data analysis, including the following elements:

- a regression equation for the reduced model

Y (ReAdmission) = 4.74 * Initial_days -1.28*Initial_admin_Emergency Admission – 0.94 * Stroke_Yes – 0.61 * Services_MRI – 0.55 *Services_CT Scan – 0.53*Gender_Nonbinary – 1.12*Anxiety_Yes – 1.22*HighBlood_Yes – 1.06*Diabetes_Yes – 1.10*Hyperlipidemia_Yes -16.87

- an interpretation of the coefficients of the reduced model

The Initial days coefficient of 4.74 means that every additional day longer stayed in the hospital will increase the likelihood of readmission by 14.43% (exp 4.74 = 114.43) compared to the relative chance of readmission than a person who has not stayed that additional day, all else equal.

Initial admin as emergency admission has a coefficient of -1.28 (exp -1.28 =0.27), indicating that a patient who is admitted due to an emergency is 0.27 times as likely to be readmitted than a person who was not, all else equal.

Stroke yes has a coefficient of -0.943 (exp -0.94 3= 0.38) which indicates that a patient who has had a stroke is 0.38 times as likely to be readmitted than a patient who had not, all else equal.

Services MRI has a coefficient of -0.617(exp -0.617=0.53), which indicates that a patient who has had an MRI is 0.53 times as likely to be readmitted than a patient who had not, all else equal.

Services CT scan has a coefficient of -0.552 (exp -0.552=0.57), which indicates that a patient who has had a CT scan is 0.57 times as likely to be readmitted than a patient who had not, all else equal.

Gender nonbinary has a coefficient of -0.533 (exp -0.533=0.58), which indicates that a nonbinary gender patient is 0.58 times as likely to be readmitted than a patient who is not, all else equal.

Anxiety yes has a coefficient of -1.12 (exp -1.12= 0.32), which indicates that a patient who has anxiety is 0.32 times as likely to be readmitted than a patient who had not, all else equal.

High blood yes has a coefficient of -1.22 (exp -1.22=0.29) which indicates that a patient who experiences high blood pressure is 0.29 times as likely to be readmitted than a patient who does not, all else equal.

Diabetes yes has a coefficient of -1.06 (exp -1.06= 0.34) indicating a patient who has diabetes is 0.34 times as likely to be readmitted than a patient who does not, all else equal.

Hyperlipidemia has a coefficient of -1.10 (exp -1.10=0.33) indicating that a patient who has hyperlipidemia is 0.33 times as likely to be readmitted than a patient who does not, all else equal.

(Jensen, 2024)

- the statistical and practical significance of the reduced model

Practically, this can be used to give a very accurate estimate of whether readmission will be necessary, which will allow for more accurate preparation in staffing, equipment and supplies. This model could also be used to replace large amounts of stored data, as the readmission category will quite accurately be predicted based on the rest of the saved fields, that data could be erased and calculated as needed for cost decrease. Some companies may also be able to determine that the columns not used in any relevant models are not worth the cost of storing long term.

- the limitations of the data analysis

All statistical models are built using past data- this model did not predict any of this data, it is created using the outcome. This means this is retrofitted, applying and fitting perfectly to past data, but we have no understanding of whether future data will follow any of the same trends. There are always unseen factors that the model cannot account for, such as natural disasters, spread of disease, or a change in the availability of some alternative form of care.

Additionally, the data itself is likely not perfect. Patients may not have a good understanding of what they are being asked, people may answer differently based on mood, recency bias, beliefs, etc. There are many potential ways datapoints could be inaccurate when analyzing a large population.

2. Recommend a course of action based on your results.

With the benefit of the reduced model, it is recommended that the hospital run a Readmission analysis on patients upon discharge from their initial stay. This will allow the hospital to then forecast staffing needs, equipment, medication, and the hospital should prepare accordingly. If desired, the hospital would also now have the option to remove the

ReAdmission field from long-term data storage and use the reduced model to determine this value when necessary.

G: Panopto Link: https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=0f379e6e-e0d7-481a-ba18-b2bc01592f31

H&I:

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